

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Computer Vision with OpenCV COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO2: *Apply image processing and feature extraction techniques, including edge detection, motion estimation, and optical flow, for analysing images.. (BL3)*

Assessment Tool Weightage: 30%

QUESTIONS: 3 Total Marks:30 Annexure A

Scenario-Based

Code of Conduct:

*I **Cheredla Jasvina /192324026** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

			Scenario based assessment			
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding ; misses key aspects	Misinterprets or fails to understand the scenario	
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues	

Appl icati on of Concep ts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts	
Decis ion Maki ng	20%	Makes well- reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided	
Clarit y of Respo nse	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganize d and difficult to understand	

Annexure A

Scenario-Based

1. Crack Detection on Concrete Surfaces Using Drone Images

Scenario: A civil inspection team is developing a computer vision system to detect thin cracks on concrete bridges and building surfaces. Images are captured using a drone under outdoor conditions. Due to changing sunlight, shadows, and camera vibration, the raw images contain uneven illumination, low contrast, and Gaussian noise.

Answers :

A. Propose a suitable image preprocessing pipeline to prepare the drone images for crack detection. Your answer should include methods for illumination correction, noise reduction, contrast enhancement, and image normalization. Justify the need for each step.

A preprocessing pipeline is a sequence of steps performed on an image before analysis. Each step improves the image quality so that cracks can be detected accurately.

Ans:

Image preprocessing improves the quality of the captured images before crack detection. Since drone images suffer from uneven lighting, shadows, blur, and noise, preprocessing is essential to make cracks more visible and improve detection accuracy.

A suitable preprocessing pipeline is:

Step 1: Image Acquisition

The drone captures high-resolution RGB images of concrete surfaces.

- Images may contain sunlight variations, shadows, dust, and vibration effects.
- Therefore, preprocessing is required before crack detection.

Step 2: Convert RGB Image to Grayscale

The captured RGB image is converted into a grayscale image.

Reason:

- Crack detection mainly depends on intensity changes rather than color.
- Grayscale reduces image complexity from three channels (Red, Green, Blue) to one channel.
- Processing becomes faster and computationally efficient.

Example:

A dark crack on a gray concrete wall is easier to detect using intensity values than RGB colors.

Step 3: Illumination Correction

Outdoor images often contain bright sunlight and dark shadow regions.

A suitable technique is:

- Adaptive Histogram Equalization (CLAHE) or
- Homomorphic Filtering

Why is it needed?

- Removes uneven illumination.
- Reduces shadow effects.
- Produces uniform brightness across the image.

Example:

Without correction:

- One part of the bridge appears bright.
- Another part is covered by shadow.

After correction:

- Both regions have similar brightness.
- Cracks become equally visible.

Step 4: Noise Reduction

Drone vibration and camera sensors introduce Gaussian noise.

A suitable filter is:

- Gaussian Filter
- Median Filter (if impulse noise is present)
- Since the scenario specifically mentions Gaussian noise, the Gaussian filter is the best choice.

Why Gaussian Filter?

- Smooths random Gaussian noise.
- Preserves the overall crack structure.
- Produces a cleaner image.

Example:

Before filtering:

Random white and black pixels appear on the concrete surface.

After filtering:

Noise disappears while crack lines remain visible.

Step 5: Contrast Enhancement

Thin cracks usually have very low contrast compared to the concrete background.

Suitable methods:

- Histogram Equalization
- CLAHE (preferred)

Why?

- Increases the difference between crack pixels and background.
- Makes thin cracks darker and easier to detect.
- Improves visibility without over-amplifying noise.

Example:

Before enhancement:

- Crack intensity = 118
- Concrete intensity = 125

Difference = only 7

After enhancement:

- Crack intensity = 70
- Concrete intensity = 180

Difference becomes much larger, making cracks easier to detect.

Step 6: Image Normalization

Normalize pixel values.

Example:

Original pixel range:

0–255

Normalized range:

0–1

or

–1 to +1

Why?

- Removes intensity variations.
- Makes all images have the same scale.
- Improves consistency for machine learning and computer vision algorithms.

Step 7: Output Image

The final image becomes:

- Uniformly illuminated
- Noise-free
- High contrast
- Normalized

This image is now ready for accurate crack detection.

Justification of Each Step

1. Grayscale Conversion

- Reduces image complexity.
- Focuses on intensity rather than color.

2. Illumination Correction

- Removes shadows and uneven lighting.
- Ensures cracks remain visible in all regions.

3. Noise Reduction

- Removes Gaussian noise caused by the drone camera.
- Prevents false crack detection.

4. Contrast Enhancement

- Highlights thin cracks.
- Improves edge visibility.

5. Image Normalization

- Standardizes intensity values.
- Improves robustness of later processing steps.

B. Select an appropriate edge detection technique for identifying thin surface cracks. Compare it with at least one other edge detection method and explain why your selected technique is more suitable for detecting fine cracks.

Ans:

The most suitable edge detection technique is the Canny Edge Detector.

Why Canny Edge Detection?

- Canny is highly effective for detecting thin surface cracks because it:
- Detects very fine edges.
- Produces continuous edge lines.
- Removes false edges caused by noise.
- Uses non-maximum suppression for accurate localization.
- Applies double thresholding to distinguish strong and weak edges.
- Uses edge tracking by hysteresis to connect weak edges that belong to real cracks.

These features make it ideal for detecting narrow, irregular crack patterns in concrete surfaces.

Comparison with Sobel Edge Detector

Sobel Edge Detector

Working:

- Calculates horizontal and vertical gradients.
- Detects edges using first-order derivatives.

Advantages:

- Simple
- Fast
- Low computational cost

Disadvantages:

- Sensitive to noise.
- Produces thick edges.

- May miss very thin cracks.
- Generates many false edges.

Canny Edge Detector

Working:

- Gaussian smoothing
- Gradient calculation
- Non-maximum suppression
- Double thresholding
- Edge tracking by hysteresis

Advantages:

- Excellent noise suppression.
- Detects thin cracks accurately.
- Produces single-pixel-wide edges.
- Better localization.
- Connects weak crack edges.

Disadvantages:

- More computationally expensive than Sobel.

Why Canny is More Suitable

Compared to Sobel, Canny is preferred because:

- It removes Gaussian noise before detecting edges.
- It detects very fine and narrow cracks.
- It produces cleaner and more continuous crack boundaries.
- It significantly reduces false edge detection.
- It performs well even under uneven outdoor lighting conditions.

Therefore, Canny Edge Detection is the most suitable technique for identifying thin concrete surface cracks.

C. After edge detection, some crack segments appear broken or disconnected. Suggest a suitable post-processing technique to connect the crack regions and improve the final detection output.

Ans:

After edge detection, some crack segments may appear broken or disconnected because of shadows, noise, or weak edges.

A suitable post-processing technique is Morphological Closing.

Morphological Closing

Morphological Closing consists of two operations:

- Dilation
- Erosion

Step 1: Dilation

- Expands crack boundaries.
- Connects nearby crack segments.
- Fills small gaps.

Step 2: Erosion

- Restores the crack to its original thickness.
- Removes unnecessary expansion caused by dilation.

Why Morphological Closing?

It helps to:

- Connect broken crack segments.
- Fill small holes inside cracks.
- Produce continuous crack structures.

- Improve the final crack detection result.

Example

Before Closing

The crack is broken into separate pieces.

After Morphological Closing

The crack becomes one continuous line.

Benefits of Morphological Closing

- Connects disconnected crack regions.
- Fills small gaps and holes.
- Produces smoother crack boundaries.
- Improves crack continuity.
- Increases the accuracy of crack measurement and analysis.

Final Conclusion

To accurately detect thin cracks in drone images of concrete surfaces:

- **Preprocessing:** Convert to grayscale → Illumination correction (CLAHE/Homomorphic Filtering) → Gaussian filtering → Contrast enhancement (CLAHE) → Image normalization.
- **Edge Detection:** Use Canny Edge Detection because it provides accurate, thin, and continuous crack edges while reducing noise.
- **Post-Processing:** Apply Morphological Closing (Dilation followed by Erosion) to connect broken crack segments and produce a complete crack detection result.

2. Human Motion Tracking in a Security Camera Video

Scenario: A security camera is installed in a hallway to monitor people moving in different directions. The system must estimate the motion of each person across video frames for tracking and activity analysis. The hallway contains plain walls, smooth floors, and occasional shadows.

Questions:

Explain how the Lucas-Kanade optical flow method can be used to estimate the motion of people between consecutive video frames.

Ans:

- A.** The Lucas–Kanade Optical Flow method is used to estimate the motion of objects by tracking how image features move between two consecutive video frames. It assumes that the brightness of a small image region remains nearly constant and that the motion between frames is small.

Working Steps

Step 1: Capture Consecutive Frames

The security camera continuously captures video frames.

Example:

- Frame 1 → Person at Position A
- Frame 2 → Person at Position B

The person's position changes because of movement.

Step 2: Detect Good Feature Points

Lucas–Kanade tracks distinctive feature points such as:

- Corners
- Edges
- Texture-rich regions

It avoids tracking plain regions because they contain little useful information.

Example:

On a person, feature points may be detected on:

- Face
- Eyes
- Shirt logo
- Bag edges
- Shoe corners

Step 3: Divide into Small Windows

A small window (for example, 5×5 or 7×7 pixels) is placed around each feature point. The algorithm assumes that all pixels inside the window move in the same direction.

Step 4: Compute Optical Flow

The algorithm compares the feature window in Frame 1 with the corresponding window in Frame 2.

It calculates:

- Horizontal motion (u)
- Vertical motion (v)

These values form the optical flow vector.

Example:

If a person moves 5 pixels to the right, the optical flow vector points toward the right.

Step 5: Repeat for All Frames

The process continues for every pair of consecutive frames.

The movement of feature points creates a motion path, allowing the system to continuously track each person.

Advantages of Lucas–Kanade

Simple and computationally efficient.

- Suitable for real-time security surveillance.
- Accurately tracks small movements.
- Works well for slow and continuous motion.

B. Discuss how the aperture effect can create errors in motion estimation, especially on uniform regions such as plain walls, floors, and clothing surfaces.

Ans:

What is the Aperture Effect?

The aperture effect occurs when only a small portion of an object is visible. In such cases, the true direction of motion cannot be determined because there is insufficient visual information.

The algorithm can estimate motion only perpendicular to an edge, not the complete motion.

How It Creates Errors

The hallway contains:

- Plain walls
- Smooth floors
- Uniform clothing surfaces

These regions have very little texture or distinctive features.

Because neighboring pixels look almost identical, the algorithm cannot determine the exact direction of movement.

As a result:

- Motion vectors become inaccurate.
- Tracking may drift.
- Some motion may not be detected correctly.

Examples

Plain Wall

A white wall has almost identical pixel values everywhere.

The algorithm cannot identify reliable feature points.

Result:

- Motion cannot be estimated accurately.

Smooth Floor

A polished floor has very little texture.

The optical flow algorithm struggles to determine movement.

Result:

- Weak or incorrect motion vectors.

Plain Clothing

If a person wears a plain black or white shirt, there are few detectable features.

Only the outline of the body may be tracked, while the actual movement inside the clothing region cannot be estimated accurately.

Effects of the Aperture Effect

- Incorrect motion estimation.
- Missing motion vectors.
- Ambiguous movement direction.
- Reduced tracking accuracy.
- Difficulty tracking people in textureless regions.

C. If a person suddenly changes speed or direction, describe how the optical flow vectors will be affected. Suggest a method to improve tracking accuracy in such situations.

Ans:

Effect on Optical Flow Vectors

Lucas–Kanade assumes that motion between consecutive frames is small and smooth.

If a person suddenly:

- Starts running,
- Stops abruptly,
- Turns quickly,

this assumption is violated.

Sudden Increase in Speed**When a person suddenly runs:**

Optical flow vectors become much longer because the displacement between frames increases.

The algorithm may fail to match the correct feature positions.

Result:

- Tracking errors.
- Loss of feature points.

Sudden Change in Direction

Suppose a person walks straight and suddenly turns left.

Before turning:

→ → → →

After turning:

↑ ↑ ↑ ↑

The optical flow vectors suddenly change direction.

The algorithm may temporarily lose the person's trajectory because it expects gradual motion.

Method to Improve Tracking Accuracy

A suitable method is the Pyramidal Lucas–Kanade Optical Flow.

How It Works

- The image is converted into multiple resolutions (image pyramid).
- Motion is first estimated at a lower resolution, where large movements appear smaller.
- The estimated motion is then refined at higher resolutions.

Advantages

- Handles large movements effectively.
- Tracks sudden speed changes.
- Improves accuracy when direction changes quickly.
- Reduces tracking failure caused by large displacements.

Other Supporting Techniques

To further improve tracking accuracy:

- Detect strong feature points using the Shi–Tomasi Corner Detector.
- Use a Kalman Filter to predict a person's future position when temporary tracking errors occur.
- Periodically re-detect feature points to replace lost ones.
- Apply shadow removal techniques to reduce errors caused by hallway shadows.

Final Conclusion

The Lucas–Kanade Optical Flow method estimates the motion of people by tracking feature points between consecutive video frames. However, the aperture effect causes errors in textureless regions such as plain walls, smooth floors, and plain clothing because there are insufficient features for accurate motion estimation. When a person suddenly changes speed or direction, the optical flow vectors change significantly, which may reduce tracking accuracy. Using the Pyramidal Lucas–Kanade Optical Flow method, along with feature detection and prediction techniques such as the Kalman Filter, improves tracking performance and ensures more reliable human motion tracking in security camera videos.

3. Vehicle Tracking at a Traffic Junction Using Video Frames

Scenario: A smart traffic monitoring system uses CCTV footage to track vehicles at a busy road junction. Vehicles move at different speeds, some stop suddenly at signals, and others turn left or right. The video also contains shadows, reflections, and partial occlusion when vehicles overlap.

Questions:

A. Explain how optical flow can be used to estimate the direction and speed of moving vehicles in the video sequence.

Ans:

Optical Flow is a computer vision technique that estimates the motion of objects by tracking the movement of pixels or feature points between consecutive video frames. In a traffic monitoring system, it helps determine the direction and speed of moving vehicles.

Working Steps

Step 1: Capture Consecutive Video Frames

The CCTV camera continuously captures video frames.

Example:

- Frame 1 → Vehicle at Position A
- Frame 2 → Vehicle at Position B

The change in position indicates that the vehicle has moved.

Step 2: Detect Feature Points

The optical flow algorithm identifies distinctive feature points on the vehicle, such as:

- Headlights
- Corners of the vehicle
- Number plate
- Wheels
- Window edges

These feature points are easier to track across frames.

Step 3: Track Feature Movement

The algorithm compares the position of each feature point in two consecutive frames.

The movement of each feature point is represented by an optical flow vector, which contains:

- Direction of movement
- Distance moved

Step 4: Estimate Direction

The direction of the optical flow vectors indicates the vehicle's movement.

Examples:

- Vectors pointing right → Vehicle moving to the right.
- Vectors pointing left → Vehicle turning left.
- Vectors pointing upward → Vehicle moving forward.
- Vectors changing direction → Vehicle is taking a turn.

Step 5: Estimate Speed

The speed is calculated from the displacement of the feature points between frames.

- Long optical flow vectors indicate high speed.
- Short optical flow vectors indicate low speed.
- Zero-length vectors indicate that the vehicle has stopped.

Example:

If a vehicle moves 15 pixels between two frames, it is moving faster than another vehicle that moves only 5 pixels.

Advantages of Optical Flow

- Tracks vehicle movement continuously.
- Estimates both speed and direction.
- Works in real-time.
- Useful for traffic monitoring and activity analysis.

B. Identify two challenges that may affect accurate motion estimation in this scenario, such as occlusion, shadows, sudden braking, or camera vibration. Explain how each challenge affects the optical flow result.

Ans:

Several factors can reduce the accuracy of optical flow. Two important challenges are occlusion and shadows.

1. Occlusion

What is Occlusion?

Occlusion occurs when one vehicle partially or completely blocks another vehicle.

Example:

A bus passes in front of a motorcycle, hiding it from the camera.

Effect on Optical Flow

- Feature points on the hidden vehicle disappear.
- The algorithm loses track of the vehicle.
- Motion vectors become incomplete or incorrect.
- Tracking may switch to the wrong vehicle when the hidden vehicle reappears.

2. Shadows

What are Shadows?

Shadows are dark regions created by vehicles or nearby objects due to sunlight or artificial lighting.

Effect on Optical Flow

- The algorithm may detect shadow boundaries as moving objects.
- False motion vectors are generated.
- Vehicle boundaries become difficult to identify.
- Tracking accuracy decreases because shadows move along with the vehicle.

Overall Effects of These Challenges

- Incorrect speed estimation.

- Wrong movement direction.
- Loss of vehicle tracking.
- False motion detection.
- Reduced tracking accuracy.

C. Suggest a suitable improvement method to make vehicle tracking more reliable when vehicles overlap or change direction suddenly.

Ans:

A suitable method to improve tracking is the Kalman Filter combined with Optical Flow.

Kalman Filter

The Kalman Filter is a prediction algorithm that estimates the future position of a moving object based on its previous motion.

How It Works

- The optical flow algorithm estimates the current position and motion of each vehicle.
- The Kalman Filter predicts the vehicle's next position.
- Even if the vehicle is temporarily hidden (occlusion) or changes direction suddenly, the predicted position helps continue tracking.
- When the vehicle becomes visible again, the prediction is corrected using the new observations.

Advantages

- Maintains tracking during temporary occlusion.
- Handles sudden braking and turning more effectively.
- Reduces tracking errors caused by missing feature points.
- Produces smoother and more stable motion trajectories.
- Improves overall vehicle tracking accuracy in busy traffic scenes.

Additional Supporting Techniques

To further improve performance:

- Use Pyramidal Lucas–Kanade Optical Flow to handle larger vehicle movements between frames.
- Apply shadow removal techniques before optical flow to reduce false motion caused by shadows.
- Detect strong feature points (such as vehicle corners and headlights) to improve tracking reliability.
- Re-detect feature points periodically if some tracked points are lost.

Final Conclusion

Optical flow estimates the direction and speed of vehicles by tracking the movement of feature points between consecutive video frames. The direction is determined by the orientation of the optical flow vectors, while the speed depends on the length of these vectors. However, challenges such as occlusion and shadows can cause incorrect motion estimation by hiding vehicles or generating false motion. To improve tracking reliability, combining Optical Flow with a Kalman Filter provides accurate prediction of vehicle positions, allowing the system to continue tracking even when vehicles overlap, stop suddenly, or change direction.